

Risk of cancer in Asian Americans

Author(s): H. Nicole Tran, Yan Li, Natalia Udaltsova, Mary Anne Armstrong, Gary D. Friedman and Arthur L. Klatsky

Source: *Cancer Causes & Control*, October 2016, Vol. 27, No. 10 (October 2016), pp. 1197-1207

Published by: Springer

Stable URL: <https://www.jstor.org/stable/10.2307/48691893>

JSTOR is a not-for-profit service that helps scholars, researchers, and students discover, use, and build upon a wide range of content in a trusted digital archive. We use information technology and tools to increase productivity and facilitate new forms of scholarship. For more information about JSTOR, please contact support@jstor.org.

Your use of the JSTOR archive indicates your acceptance of the Terms & Conditions of Use, available at <https://about.jstor.org/terms>



JSTOR

Springer is collaborating with JSTOR to digitize, preserve and extend access to *Cancer Causes & Control*

Risk of cancer in Asian Americans: a Kaiser Permanente cohort study

H. Nicole Tran¹ · Yan Li² · Natalia Udaltsova³ · Mary Anne Armstrong³ · Gary D. Friedman³ · Arthur L. Klatsky³

Received: 1 December 2015 / Accepted: 10 August 2016 / Published online: 25 August 2016
© Springer International Publishing Switzerland 2016

Abstract

Purpose To supplement published cohort data about incident cancer in Asian Americans (Asians) including risk of specific Asian ethnic groups.

Methods A cohort study in 124,193 persons (13,344 Asians) with baseline examination data in 1978–1985 used Cox proportional hazards models with seven covariates to estimate hazard ratios (HRs) and 95 % confidence intervals (CIs).

Results Through 2012 cancer was diagnosed in 18,687 persons including 1,522 Asians. Compared to Whites, the HR (CIs) for any cancer in Asians was 0.8 (0.7–0.9, $p < 0.001$). Lower Asian risk was stronger for men (HR = 0.7, $p < 0.001$) than for women (HR = 0.9, $p = 0.003$). Lower Asian vs. White risks with $p < 0.05$ were found for cancers of the upper airway digestive area, hematologic malignancies, melanoma, and cancers of the prostate, bladder, and brain. Melanoma contributed substantially to lower Asian risk, especially in women. HRs for specific Asian groups versus Whites follow: Chinese = 0.9 ($p < 0.001$), Japanese = 0.9 ($p = 0.01$), Filipinos = 0.8 ($p < 0.001$), South Asians = 0.5 ($p < 0.001$), and Other Asians = 0.7 ($p = 0.006$). Both South Asian men and women had lower risk than Whites, and South Asians had lower risk than any other racial/ethnic group.

Conclusions Asians had lower cancer risk than Whites, due to lower risk of several cancer types. Each Asian ethnic group had lower risk than Whites with South Asians at the lowest risk.

Keywords Epidemiology · Risk factors · Ethnicity · Asian Americans · South Asians

Introduction

Asian Americans (Asians) have rapidly increased in numbers and prominence in the USA. By 2014 their numbers had reached 20,250,250 or 6.3 % of the total US population [1]. However, there are relatively few epidemiological studies of Asians in general and of comprehensive cancer incidence in particular. Prospective data with control for risk factors about cancer risk of specific Asian ethnic groups are especially sparse [2–6]. Asians are a heterogeneous race consisting of diverse ethnic subgroups, but most published studies regard the Asian population as one homogenous group.

Available data about Asian cancer rates in comparison with that of Whites suggest the following: (1) overall lower risk of cancer incidence and/or mortality [6–8], (2) decreased risk of melanoma [9, 10], (3) decreased risk of hematologic malignancies [11, 12], (4) increased risk of stomach cancer [8, 13–15], liver cancer [8, 14–16], cervical cancer [3, 4, 6, 17], and lung adenocarcinoma in non-smoking women [18]. Hints of cancer risk disparities between Asian subgroups include especially high risk of stomach and liver cancer in Japanese, Koreans, and Chinese [6, 13, 14] and of cervical cancer in Southeast Asian women, especially Vietnamese women [3, 4, 17, 19].

✉ H. Nicole Tran
Nicole.H.Tran@kp.org

¹ Department of Internal Medicine, Kaiser Permanente Medical Center, 275 W MacArthur Blvd, Oakland, CA 94611, USA

² Department of Hematology and Oncology, Kaiser Permanente Medical Center, Oakland, CA, USA

³ Division of Research, Kaiser Permanente Northern California, Oakland, CA, USA

Previously, we have been using an available large multi-ethnic free-living cohort to study relationships of Asian ethnicity to several cardiovascular and lung disease outcomes. We have performed and report here an analysis of cancer risk. We hope this will lead to greater understanding of cancer etiology and appropriate changes in screening and patient education practices.

Materials and methods

Study population and data

The study protocols were approved by the Kaiser Permanente Northern California Institutional Review Board. The examinees were a multi-ethnic [20] cohort of 124,193 men and women free at baseline of cancer history who were members of a comprehensive prepaid health plan in the San Francisco Bay Area. Members are believed to reflect the full socioeconomic spectrum of the general population except for the extremes of income [21]. Baseline data were obtained from health plan records and from research questionnaires at health examinations from 1978 through 1985. Usually taken as a voluntary routine health appraisal, the examination [22] included health measurements and queries about socio-demographic status, habits, medical history, and symptoms. The study cohort comprised 79.8 % of all examinees; the remainder included persons who took the examination during absences of a special research clerk and persons who declined, partially because of lack of fluency in English.

The questionnaire item “What is your race?” identified White persons, African Americans (Blacks), Asian Americans (Asians), Hispanics, and Others. Asians were self-identified as Chinese, Filipinos, Japanese, and Other Asians (largely Korean or Vietnamese). South Asians, defined as persons of Indian, Pakistani, Bangladeshi, or Sri Lankan ancestry, was not a checklist category. We identified South Asians from lists of Other Asians and Others that included birthplace, religion, names, marital status, and, in some instances, birthplace of parents and grandparents. All data so used were supplied on the examination questionnaires, and determination of South Asian ethnicity was made independently by two investigators (MAA and ALK). The almost complete agreement of these investigators led them to conclude that specificity of the South Asian determination was virtually 100 %. With respect to sensitivity, South Asian persons would not be screened in the manner described only if the individual self-classified as White, Black, Chinese, Japanese, Filipino, or Hispanic. Such self-misclassification seems unlikely except as a mistake. Most South Asians were Asian Indians. The proportions of Asian

subgroups born in the USA varied substantially from 7 % of Filipinos to 79 % of Japanese.

Cancer subjects

Occurrence of cancer ($n = 18,637$) was ascertained from the Kaiser Permanente Cancer Registry, which covers all subscribers and contributes to the local Surveillance, Epidemiology, and End Results (SEER) program. Endpoints studied included any cancer and multiple specific cancer types. We report here data about the 14 commonest types in men and the 16 commonest types in women. These types included 2 multi-cancer composites: (1) Upper airway digestive (UAD) cancers included 552 subjects with one or more cancers of the lip, tongue, gum, mouth, oropharynx, hypopharynx, unspecified oral cavity, esophagus, and larynx. (2) Hematologic malignancies included 1,639 subjects with one or more of Hodgkin’s disease, non-Hodgkin’s lymphoma, multiple myeloma, lymphocytic leukemia, myelocytic leukemia, and other leukemia.

Table 1 presents selected distributions of the study population and cancer subjects in the study population by sex and race/ethnicity.

Analytic methods

Subjects were followed until 31 December 2012, diagnosis of cancer, death, or other termination of health plan membership. Mean follow-up was 17.8 years, yielding an estimated 2,216,631 person years of follow-up. Mean follow-up times for selected specific race ethnic groups were: Whites = 17.9 years, All Asians = 19.4 years, Chinese = 19.3 years, Japanese = 19.0 years, Filipinos = 20.3 years, South Asians = 19.0 years, and Other Asians = 18.7 years. Multivariate models used the Cox proportional hazards model. Models for all persons and for subgroup strata included as covariates age (continuous), education (no college referent, some college, college graduate), body mass index ($<25 \text{ kg/m}^2$ referent, $25\text{--}29 \text{ kg/m}^2$, $\geq 30 \text{ kg/m}^2$), marital status (married referent, never married, formerly married), cigarette smoking (never smoked referent, ex-smoker, current <1 pack per day, current ≥ 1 pack per day), and alcohol intake (lifelong abstainers referent, ex-drinkers, <1 drink per day, $1\text{--}2$ drinks per day, ≥ 3 drinks per day).

Race/ethnicity was studied for all endpoints as follows: (1) models in all persons with Whites, the largest racial group, as referent, plus Blacks, all Asians, Hispanics, Others, (2) similar models in all persons with Whites as referent plus Blacks, Hispanics, and the Asian subgroups (Chinese, Filipinos, Japanese, South Asians, and Other Asians), (3) models in Asians only with Chinese, the largest Asian group, as referent plus other Asian ethnicities

Table 1 Study group and cancer subjects

Group	<i>n</i> (%) subjects	Mean baseline age	<i>n</i> (%) cancer	Rate/1000 person-yrs
Total	124,193 (100)	40.6	18,637 (100)	8.4
Men	55,040 (44.3)	41.4	8,553 (47.5)	9.2
Women	69,153 (55.7)	40.0	9,784 (52.5)	7.8
White	68,597 (55.2)	42.7	11,072 (59.4)	9.5
Men	32,152 (25.9)	42.8	5,374 (28.8)	10.0
Women	36,445 (29.3)	42.6	5,698 (30.6)	9.1
Black	33,625 (27.1)	38.4	5,131 (27.5)	8.1
Men	13,193 (10.6)	40.1	2,417 (13.0)	10.2
Women	20,432 (16.5)	37.3	2,714 (14.6)	6.8
All Asian	13,344 (10.7)	38.4	1,522 (8.2)	5.9
Men	5,910 (4.8)	39.5	662 (3.6)	5.8
Women	7,434 (6.0)	37.4	860 (4.6)	6.0
Chinese	5,910 (4.8)	39.0	728 (3.9)	6.4
Men	2,738 (2.2)	40.1	338 (1.8)	6.4
Women	3,172 (2.6)	38.0	390 (2.1)	6.4
Japanese	1,653 (1.2)	41.4	247 (1.3)	7.9
Men	673 (0.5)	42.4	247 (0.5)	7.5
Women	980 (0.8)	40.8	152 (0.8)	8.1
Filipino	4,223 (3.4)	37.7	447 (2.4)	5.2
Men	1,732 (1.4)	39.1	175 (0.9)	4.9
Women	2,491 (2.0)	36.8	272 (1.5)	5.4
South Asian	689 (0.6)	34.3	38 (0.2)	2.9
Men	399 (0.3)	35.7	22 (0.1)	2.9
Women	290 (0.2)	32.3	16 (0.1)	2.8
Other Asian	869 (0.7)	34.7	62 (0.3)	4.3
Men	368 (0.3)	35.5	32 (0.2)	5.3
Women	501 (0.4)	34.0	30 (0.2)	3.5
Hispanic	5,620 (4.5)	36.7	614 (3.3)	5.7
Men	2,521 (2.0)	37.2	277 (1.5)	5.8
Women	3,099 (2.5)	36.3	337 (1.8)	5.6
Other/Misc	3,007 (2.4 %)	37.8	298 (1.6)	5.9
Men	1,264 (1.0)	39.7	135 (0.7)	6.1
Women	1,743 (1.4)	39.1	163 (0.9)	5.7

We report hazard ratios (HRs), 95 % confidence intervals (CIs), and associated *p* values.

Results

Any cancer

The Cox proportional hazards models presented in Table 2 show that Asians had an approximately 20 % lower risk than Whites of developing any cancer, with the reduced risk greater in men (~30 %) than in women (~10 %). South Asians were at the lowest risk in each sex (~40 % lower in women and ~50 % lower in men). The magnitude of the lower risk of cancer was less but generally

similar among the other specific Asian groups. Excluding melanoma from the model for any cancer attenuated the Asians vs. Whites HRs for all persons to 0.9 ($p < 0.001$), for men to 0.8 ($p < 0.001$), and for women to 0.95 ($p = 0.02$). Excluding melanoma also lessened the reduced risk for specific Asian groups, with a HR of 0.93 for Chinese ($p = 0.07$), 0.94 Japanese ($p = 0.3$), 0.86 for Filipinos ($p = 0.003$), and 0.58 for South Asians ($p < 0.001$). Models omitting other cancer types, one at a time had little effect on the Asian/White HRs; e.g., minus lung cancer HR = 0.78, minus colorectal cancer HR = 0.79, minus bladder cancer HR = 0.82, minus liver cancer HR = 0.79, minus prostate cancer in men HR = 0.73, and minus breast cancer in women = 0.79. In a model with Blacks as referent the Asian/Black HR was 0.8 ($p < 0.001$) but this was

Table 2 Adjusted^a hazard ratios for any cancer by race/ethnicity

Race/ethnicity	All persons	All men	All women
White (referent)	1.0 (–)	1.0 (–)	1.0 (–)
Black	1.1 (1.0–1.1) [$p < 0.001$]	1.2 (1.2–1.3) [$p < 0.001$]	0.9 (0.6–0.9) [$p < 0.001$]
Hispanic	0.8 (0.7–0.9) [$p < 0.001$]	0.8 (0.7–0.9) [$p < 0.001$]	0.8 (0.7–0.9) [$p < 0.001$]
All Asians	0.8 (0.8–0.9) [$p < 0.001$]	0.7 (0.6–0.8) [$p < 0.001$]	0.9 (0.8–1.0) [$p = 0.001$]
Chinese	0.9 (0.8–0.9) [$p < 0.001$]	0.8 (0.6–0.9) [$p < 0.001$]	0.9 (0.8–1.0) [$p = 0.2$]
Japanese	0.9 (0.8–1.0) [$p = 0.010$]	0.7 (0.6–0.9) [$p < 0.001$]	1.0 (0.8–1.1) [$p = 0.6$]
Filipino	0.8 (0.7–0.9) [$p < 0.001$]	0.7 (0.6–0.8) [$p < 0.001$]	0.9 (0.8–1.0) [$p < 0.03$]
South Asian	0.5 (0.4–0.7) [$p < 0.001$]	0.5 (0.3–0.7) [$p = 0.002$]	0.6 (0.3–0.9) [$p = 0.02$]
Other Asian	0.7 (0.6–0.9) [$p = 0.006$]	0.9 (0.6–1.2) [$p = 0.4$]	0.6 (0.4–0.8) [$p < 0.004$]
Other/Misc [†]	0.9 (0.8–1.0) [$p = 0.07$]	0.9 (0.8–1.1) [$p = 0.3$]	0.9 (0.8–1.0) [$p = 0.08$]

^a Hazard ratios (95 % CIs) from Cox models with age, sex, race, smoking, alcohol, BMI, education, and marital status (vs Whites as referent)

[†] Other than White, Black, Asian, Hispanic

concentrated in men (HR = 0.6, $p < 0.001$) and absent in women (HR = 1.0). With Hispanics as referent, the Asian HRs were 1.0 for all persons, 0.9 for men ($p = 0.09$), and 1.1 for women ($p = 0.09$).

Cancer types with Asians at higher risk than Whites

Tables 3 and 4 present data for women and men, respectively. Both Asian men and women were at higher risk of stomach cancer with increased risk concentrated in Japanese. Asian men had increased liver cancer risk, strongest in Chinese. For the sexes combined the p value of the higher Asian risk was <0.001 for both liver (HR = 2.2 [1.6–3.2]) and stomach cancer (2.8 [2.2–3.5]). Asian inter-ethnic comparisons for stomach cancer showed several disparities, with Japanese higher than Filipinos (HR = 4.1, $p = 0.01$) and Other Asians higher than Chinese (HR = 3.3, $p = 0.02$) or Filipinos (HR = 6.4, $p = 0.004$). There were no substantial Asian inter-ethnic disparities for liver cancer.

Asian women were at higher risk of invasive cancer of the cervix (Table 3), with risk highest among Japanese and Filipino women and no cases in South Asian women. For Other Asians, which included Vietnamese and Korean, the HR was 2.4 (0.2–18.3). There were no substantial Asian inter-ethnic disparities for this cancer type. We also studied 631 cases of noninvasive cervical cancer; among these the Asian vs White HR was 0.9 (0.6–1.2).

Asian women, especially Chinese and Filipinos, were at higher risk of lung cancer (Table 3), but Asian men did not have higher lung cancer risk than Whites. With the sexes combined the Asian/White HR was 1.1 (1.0–1.4, $p = 0.1$).

Also noteworthy were higher risk of South Asian men for UAD cancer and higher risk of ovarian cancer in South Asian women; these data were based upon small numbers with wide CIs for the estimates.

Cancer types with Asians at lower risk than Whites

Asians had lower ($p < 0.05$) risk of six cancer types (Tables 3 and 4): (1) Asians of each sex and of each Asian ethnicity had ~90 % lower risk of melanoma than Whites, (2) Asian men had lower risk than Whites for UAD, and (3) Asian men also had lower bladder cancer risk with $p < 0.001$. (4) Asian women and men were similar in having slight reductions of brain cancer risk (Tables 3 and 4) with $p < 0.05$ for the sexes combined. (5) Asians of each sex had substantially lower risk of hematologic malignancies. (6) Asian men had ~30 % lower risk than White men of prostate cancer.

The decreased risk in Asian men of the genitourinary cancers (prostate and bladder) cut across all ethnic subgroups (Table 4); for prostate cancer it was most pronounced in South Asian men (50 % lower than Whites), and for bladder cancer it was most pronounced in Filipinos (60 % lower risk). Reduced risk of bladder cancer was similar for invasive (HR = 0.5) and noninvasive neoplasms (HR = 0.6).

The lower Asian vs. White risk of hematologic malignancies also appeared to cut across all Asian ethnicities. Within this group of malignancies, data for the sexes combined showed that Asians were at lower risk of non-Hodgkin's lymphoma (HR = 0.7 [0.5–0.9]; $p = 0.007$), lymphocytic leukemia (HR = 0.3 [0.1–0.8]; $p < 0.001$), and myelocytic leukemia (HR = 0.5 [0.2–1.0]; $p = 0.06$), but not of multiple myeloma (HR = 1.1 [0.7–1.7]). For lymphocytic leukemia, lower risk was concentrated in chronic lymphocytic leukemia (HR = 0.2 [0.1–0.7]; $p < 0.001$).

Cancer types for which Asians and Whites had similar risk

There was no overall difference between Asians and Whites for risk of colorectal, lung, pancreatic, kidney, thyroid, and female breast cancer.

Table 3 Adjusted^a hazard ratios in women for selected cancer types for all Asians and specific Asian ethnicities (versus Whites)

Cancer type (ICD-9) (Total <i>n</i> in women)	Asians HR (95 % CI) <i>n</i> subjects [<i>p</i> value]	Chinese	Japanese	Filipino	South Asians	Other Asians
UAD (140-50, [except -42, 47], 161) (<i>n</i> = 55)	0.6 (0.3–1.3) <i>n</i> = 7 (<i>p</i> = 0.2)	0.6 (0.2–1.7) <i>n</i> = 3 (<i>p</i> = 0.3)	0.8 (0.2–3.4) <i>n</i> = 2 (<i>p</i> = 0.8)	0.6 (0.1–2.6) <i>n</i> = 2 (<i>p</i> = 0.5)	–	–
Stomach (151) (<i>n</i> = 134)	2.0 (1.0–3.9) <i>n</i> = 13 (<i>p</i> = 0.04)	1.5 (0.6–3.9) <i>n</i> = 5 (<i>p</i> = 0.4)	4.1 (1.6–10.5) <i>n</i> = 5 (<i>p</i> = 0.003)	1.1 (0.2–4.5) <i>n</i> = 2 (<i>p</i> = 0.9)	<i>n</i> = 0	3.2 (0.4–24.0) <i>n</i> = 1 (<i>p</i> = 0.3)
Colorectal (153-4) (<i>n</i> = 1,059)	1.0 (0.7–1.2) <i>n</i> = 81 (<i>p</i> = 0.7)	1.2 (0.9–1.6) <i>n</i> = 46 (<i>p</i> = 0.3)	1.1 (0.6–1.7) <i>n</i> = 16 (<i>p</i> = 0.8)	0.6 (0.5–1.0) <i>n</i> = 15 (<i>p</i> = 0.04)	0.5 (0.1–3.6) <i>n</i> = 1 (<i>p</i> = 0.5)	0.7 (0.2–2.3) <i>n</i> = 3 (<i>p</i> = 0.6)
Liver (155) (<i>n</i> = 76)	1.5 (0.7–3.2) <i>n</i> = 10 (<i>p</i> = 0.1)	1.7 (0.6–4.7) <i>n</i> = 5 (<i>p</i> = 0.3)	0.9 (0.1–6.6) <i>n</i> = 1 (<i>p</i> = 0.9)	1.0 (0.2–4.6) <i>n</i> = 2 (<i>p</i> = 0.96)	–	5.5 (1.3–24.4) <i>n</i> = 2 (<i>p</i> = 0.02)
Pancreas (157) (<i>n</i> = 278)	1.0 (0.6–1.6) <i>n</i> = 19 (<i>p</i> = 0.8)	1.0 (0.5–2.0) <i>n</i> = 9 (<i>p</i> = 0.9)	1.5 (0.7–3.4) <i>n</i> = 6 (<i>p</i> = 0.3)	0.7 (0.3–2.0) <i>n</i> = 4 (<i>p</i> = 0.5)	–	–
Lung (162) (<i>n</i> = 966)	1.4 (1.1–1.8) <i>n</i> = 76 (<i>p</i> = 0.007)	1.6 (1.1–2.2) <i>n</i> = 35 (<i>p</i> = 0.02)	1.0 (0.6–1.8) <i>n</i> = 13 (<i>p</i> = 0.96)	2.3 (1.5–3.4) <i>n</i> = 28 (<i>p</i> < 0.001)	<i>n</i> = 0	–
Kidney (189) (<i>n</i> = 162)	1.0 (0.5–2.1) <i>n</i> = 10 (<i>p</i> = 0.98)	1.0 (0.4–2.9) <i>n</i> = 4 (<i>p</i> = 0.97)	0.6 (0.1–4.3) <i>n</i> = 1 (<i>p</i> = 0.6)	1.0 (0.3–3.4) <i>n</i> = 3 (<i>p</i> = 0.96)	–	3.8 (0.9–16.2) <i>n</i> = 2 (<i>p</i> = 0.06)
Bladder (188) (<i>n</i> = 222)	0.9 (0.5–1.5) <i>n</i> = 14 (<i>p</i> = 0.6)	1.4 (0.7–2.7) <i>n</i> = 10 (<i>p</i> = 0.4)	–	1.0 (0.4–2.8) <i>n</i> = 4 (<i>p</i> = 0.99)	–	–
Breast (174) (<i>n</i> = 3,639)	1.1 (0.9–1.2) <i>n</i> = 383 (<i>p</i> = 0.4)	1.1 (1.0–1.3) <i>n</i> = 175 (<i>p</i> = 0.1)	1.0 (0.8–1.3) <i>n</i> = 60 (<i>p</i> = 0.8)	1.0 (0.9–1.3) <i>n</i> = 127 (<i>p</i> = 0.6)	0.8 (0.4–1.6) <i>n</i> = 9 (<i>p</i> = 0.5)	0.7 (0.4–1.2) <i>n</i> = 12 (<i>p</i> = 0.2)
Cervix-invasive (180) (<i>n</i> = 96)	2.4 (1.3–4.6) <i>n</i> = 18 (<i>p</i> < 0.001)	1.9 (0.8–4.8) <i>n</i> = 6 (<i>p</i> = 0.2)	3.5 (1.1–11.6) <i>n</i> = 3 (<i>p</i> = 0.04)	3.2 (1.4–7.7) <i>n</i> = 8 (<i>p</i> = 0.006)	–	2.4 (0.3–18.3) <i>n</i> = 1 (<i>p</i> = 0.4)
Ovary (183) (<i>n</i> = 341)	1.0 (0.7–1.5) <i>n</i> = 35 (<i>p</i> = 0.8)	0.8 (0.4–1.5) <i>n</i> = 11 (<i>p</i> = 0.4)	1.3 (0.6–2.7) <i>n</i> = 3 (<i>p</i> = 0.8)	1.5 (0.8–2.6) <i>n</i> = 14 (<i>p</i> = 0.2)	3.6 (1.1–11.6) <i>n</i> = 3 (<i>p</i> = 0.03)	–
Uterus (182) (<i>n</i> = 688)	0.8 (0.6–1.1) <i>n</i> = 65 (<i>p</i> = 0.2)	0.9 (0.6–1.3) <i>n</i> = 29 (<i>p</i> = 0.5)	1.3 (0.8–2.2) <i>n</i> = 15 (<i>p</i> = 0.3)	0.7 (0.4–1.1) <i>n</i> = 18 (<i>p</i> = 0.8)	0.8 (0.2–3.3) <i>n</i> = 2 (<i>p</i> = 0.1)	0.3 (0.04–1.8) <i>n</i> = 1 (<i>p</i> = 0.3)
Brain (191) (<i>n</i> = 80)	0.4 (0.2–1.3) <i>n</i> = 4 (<i>p</i> = 0.3)	0.7 (0.2–2.3) <i>n</i> = 3 (<i>p</i> = 0.5)	–	0.3 (0.0–2.3) <i>n</i> = 1 (<i>p</i> = 0.2)	–	–
Thyroid (193) (<i>n</i> = 118)	1.1 (0.7–2.0) <i>n</i> = 21 (<i>p</i> = 0.7)	1.5 (0.8–2.9) <i>n</i> = 12 (<i>p</i> = 0.2)	1.3 (0.4–4.3) <i>n</i> = 3 (<i>p</i> = 0.6)	0.8 (0.3–1.9) <i>n</i> = 6 (<i>p</i> = 0.6)	<i>n</i> = 0	<i>n</i> = 0
Hematologic (200-208) (<i>n</i> = 727)	0.6 (0.4–0.8) <i>n</i> = 43 (<i>p</i> < 0.001)	0.7 (0.5–1.0) <i>n</i> = 16 (<i>p</i> = 0.004)	0.9 (0.5–1.6) <i>n</i> = 11 (<i>p</i> = 0.7)	0.5 (0.3–0.9) <i>n</i> = 12 (<i>p</i> = 0.02)	–	1.0 (0.4–1.9) <i>n</i> = 4 (<i>p</i> = 0.9)
Melanoma (172) (<i>n</i> = 514)	0.1 (0.1–0.2) <i>n</i> = 10 (<i>p</i> < 0.001)	0.2 (0.1–0.4) <i>n</i> = 7 (<i>p</i> < 0.001)	0.1 (0.0–0.5) <i>n</i> = 1 (<i>p</i> < 0.001)	0.1 (0.0–0.3) <i>n</i> = 2 (<i>p</i> < 0.001)	–	–

Table 3 continued

Cancer type (ICD-9) (Total <i>n</i> in women)	Asians HR (95 % CI) <i>n</i> subjects [<i>p</i> value]	Chinese	Japanese	Filipino	South Asians	Other Asians
All, except Melanoma (<i>n</i> = 9,386)	1.0 (0.9–1.0) <i>n</i> = 870 (<i>p</i> = 0.2)	1.0 (0.9–1.1) <i>n</i> = 383 (<i>p</i> = 0.99)	1.0 (0.9–1.2) <i>n</i> = 151 (<i>p</i> = 0.7)	0.9 (0.8–1.1) <i>n</i> = 270 (<i>p</i> = 0.4)	0.6 (0.4–1.0) <i>n</i> = 16 (<i>p</i> = 0.06)	0.6 (0.4–0.9) <i>n</i> = 30 (<i>p</i> = 0.02)

^a Cox models with age, race, smoking, alcohol, body mass index (BMI), education, and marital status (versus Whites as referent). Separate model for Asians and specific Asian subsets ICD-9+ International Classification of Diseases, 9th Revision. UAD upper airway digestive. HR hazard ratio. CI 95 % confidence interval. *n* number

Covariate relationships

Table 5 presents selected baseline traits in Whites and Asian groups, and Table 6 presents adjusted relationships of these traits to cancer risk. White men had higher all-cancer risk than White women, but the reverse was the case in Asians. Asian men were at higher risk than Asian women for several of the non-sex-related cancers, including those of the UAD (HR = 2.6), stomach (HR = 2.8), liver (HR = 2.2), lung (HR = 1.2), large intestine (HR = 1.2), and hematologic system (HR = 1.7), all with $p < 0.001$. Other covariate relationships were mostly similar for Asians and Whites: Increased age, BMI, alcohol, and smoking were associated with higher risk of cancer, while college graduation and marital status were not related.

Discussion

Cancer types with Asians at higher risk than Whites

These data confirm the previously reported higher risk of Asians for liver, stomach, and invasive cervical cancer [6, 8, 14–16, 23, 24]. Environmental exposures are likely factors for all three. High prevalence of chronic viral hepatitis B infection in Asians is an accepted explanation for the high liver cancer risk [7, 14–16, 25]. Mechanisms for the increased stomach cancer risk are less clear, but could include differences in prevalence of *Helicobacter pylori* infection [16, 26]. It is noteworthy that Filipinos did not share the increased stomach cancer risk; a previous report [27] indicates that a low gastric cancer incidence in Filipinos may be explained by high prevalence of a less pathogenetic strain of *Helicobacter pylori* in the Philippines. In the present analysis, high risk of invasive cervical cancer was found in all Asian ethnic groups. It is noteworthy that this was not found for noninvasive cervical disease. Delayed diagnosis in Asian women is a possible explanation for the increased risk of cervical cancers, which has been attributed to decreased screening rates for cervical, breast, and colorectal cancer [3, 13]: For cervical cancer screening, this has been attributed to cultural views on Pap screening [17].

Established risk factors for cervical cancer are HPV infection, smoking, the number and circumcision status of partners, and participation in screening, the usual method of diagnosis. Of these, only smoking was controlled in our analyses.

Cancer types with Asians at lower risk than Whites

The very low risk of melanoma in all Asian ethnic groups was expected and is presumably genetically mediated via

Table 4 Adjusted^a hazard ratios in men for selected cancer types for all Asians and specific Asian ethnicities (versus Whites)

Cancer type (ICD-9) (Total <i>n</i> in men)	Asians HR (95 % CI) <i>n</i> subjects [<i>p</i> value]	Chinese	Japanese	Filipino	South Asians	Other Asians
UAD (140–50, [except 142, 7], 161) (<i>n</i> = 397)	0.5 (0.3–1.0) <i>n</i> = 20 (<i>p</i> = 0.02)	0.5 (0.3–1.1) <i>n</i> = 8 (<i>p</i> = 0.08)	0.4 (0.1–1.6) <i>n</i> = 2 (<i>p</i> = 0.2)	0.5 (0.2–1.1) <i>n</i> = 2 (<i>p</i> = 0.08)	3.5 (1.3–9.6) <i>n</i> = 4 (<i>p</i> = 0.07)	– <i>n</i> = 0
Stomach (151) (<i>n</i> = 269)	1.7 (1.1–2.6) <i>n</i> = 30 (<i>p</i> = 0.01)	1.8 (1.0–3.1) <i>n</i> = 16 (<i>p</i> = 0.03)	2.6 (1.2–5.7) <i>n</i> = 7 (<i>p</i> = 0.01)	0.6 (0.2–1.8) <i>n</i> = 3 (<i>p</i> = 0.4)	– <i>n</i> = 0	6.7 (0.4–18.0) <i>n</i> = 4 (<i>p</i> < 0.001)
Colorectal (153.4) (<i>n</i> = 973)	1.0 (0.8–1.3) <i>n</i> = 84 (<i>p</i> = 0.96)	1.0 (0.7–1.4) <i>n</i> = 43 (<i>p</i> = 0.9)	1.2 (0.7–2.0) <i>n</i> = 15 (<i>p</i> = 0.5)	0.9 (0.6–1.4) <i>n</i> = 22 (<i>p</i> = 0.7)	0.3 (0.1–2.0) <i>n</i> = 1 (<i>p</i> = 0.2)	1.0 (0.3–3.0) <i>n</i> = 3 (<i>p</i> = 0.95)
Liver (155) (<i>n</i> = 137)	5.5 (3.5–8.8) <i>n</i> = 35 (<i>p</i> < 0.001)	7.9 (4.6–13.7) <i>n</i> = 22 (<i>p</i> < 0.001)	3.1 (0.9–9.9) <i>n</i> = 3 (<i>p</i> = 0.06)	3.8 (1.7–8.6) <i>n</i> = 7 (<i>p</i> = 0.002)	– <i>n</i> = 0	11.2 (3.4–36.6) <i>n</i> = 3 (<i>p</i> < 0.001)
Pancreas (157) (<i>n</i> = 257)	0.9 (0.6–1.4) <i>n</i> = 19 (<i>p</i> = 0.6)	0.9 (0.5–1.6) <i>n</i> = 9 (<i>p</i> = 0.6)	1.4 (0.6–3.5) <i>n</i> = 6 (<i>p</i> = 0.4)	0.4 (0.2–1.3) <i>n</i> = 4 (<i>p</i> = 0.1)	0.9 (0.1–6.4) <i>n</i> = 1 (<i>p</i> = 0.9)	1.0 (0.5–8.1) <i>n</i> = 2 (<i>p</i> = 0.3)
Lung (162) (<i>n</i> = 1,023)	1.0 (0.8–1.3) <i>n</i> = 76 (<i>p</i> = 0.9)	1.1 (0.8–1.6) <i>n</i> = 46 (<i>p</i> = 0.5)	0.8 (0.4–1.5) <i>n</i> = 9 (<i>p</i> = 0.4)	0.9 (0.5–1.3) <i>n</i> = 21 (<i>p</i> = 0.5)	– <i>n</i> = 0	1.2 (0.5–3.3) <i>n</i> = 4 (<i>p</i> = 0.7)
Kidney (189) (<i>n</i> = 221)	1.0 (0.7–1.7) <i>n</i> = 23 (<i>p</i> = 0.6)	0.6 (0.3–1.4) <i>n</i> = 6 (<i>p</i> = 0.2)	1.5 (0.5–3.9) <i>n</i> = 4 (<i>p</i> = 0.5)	1.5 (0.8–3.0) <i>n</i> = 10 (<i>p</i> = 0.2)	1.8 (0.3–7.3) <i>n</i> = 2 (<i>p</i> = 0.4)	1.2 (0.2–8.3) <i>n</i> = 1 (<i>p</i> = 0.9)
Bladder (188) (<i>n</i> = 591)	0.5 (0.3–0.7) <i>n</i> = 31 (<i>p</i> < 0.001)	0.6 (0.3–0.9) <i>n</i> = 18 (<i>p</i> = 0.01)	0.6 (0.3–1.3) <i>n</i> = 6 (<i>p</i> = 0.2)	0.3 (0.1–0.6) <i>n</i> = 5 (<i>p</i> = 0.003)	0.9 (0.2–3.5) <i>n</i> = 2 (<i>p</i> = 0.8)	– <i>n</i> = 0
Prostate (185) (<i>n</i> = 3,408)	0.7 (0.6–0.8) <i>n</i> = 230 (<i>p</i> < .001)	0.7 (0.6–0.9) <i>n</i> = 116 (<i>p</i> = 0.001)	0.6 (0.4–0.9) <i>n</i> = 30 (<i>p</i> = 0.01)	0.7 (0.5–0.9) <i>n</i> = 64 (<i>p</i> = 0.003)	0.5 (0.2–0.9) <i>n</i> = 8 (<i>p</i> = 0.03)	0.9 (0.5–1.6) <i>n</i> = 12 (<i>p</i> = 0.7)
Brain (191) (<i>n</i> = 106)	0.5 (0.3–1.2) <i>n</i> = 8 (<i>p</i> = 0.1)	0.8 (0.3–2.0) <i>n</i> = 5 (<i>p</i> = 0.98)	1.0 (0.2–4.2) <i>n</i> = 2 (<i>p</i> = 0.98)	0.2 (0.0–1.6) <i>n</i> = 1 (<i>p</i> = 0.2)	– <i>n</i> = 0	– <i>n</i> = 0
Thyroid (183) (<i>n</i> = 54)	1.3 (0.6–2.8) <i>n</i> = 8 (<i>p</i> = 0.6)	0.7 (0.2–2.9) <i>n</i> = 2 (<i>p</i> = 0.6)	2.5 (0.6–10.5) <i>n</i> = 2 (<i>p</i> = 0.2)	1.3 (0.4–4.5) <i>n</i> = 3 (<i>p</i> = 0.7)	– <i>n</i> = 0	2.9 (0.4–21.5) <i>n</i> = 1 (<i>p</i> = 0.3)
Hematologic (200–208) (<i>n</i> = 887)	0.7 (0.6–0.9) <i>n</i> = 70 (<i>p</i> = 0.009)	0.7 (0.5–1.0) <i>n</i> = 35 (<i>p</i> = 0.08)	0.7 (0.3–1.2) <i>n</i> = 9 (<i>p</i> = 0.2)	0.7 (0.4–1.1) <i>n</i> = 18 (<i>p</i> = 0.09)	0.9 (0.3–2.3) <i>n</i> = 4 (<i>p</i> = 0.8)	1.0 (0.4–2.8) <i>n</i> = 4 (<i>p</i> = 0.96)
Melanoma (172) (<i>n</i> = 650)	0.03 (0.01–0.1) <i>n</i> = 4 (<i>p</i> < 0.001)	0.03 (0.01–0.2) <i>n</i> = 2 (<i>p</i> < 0.001)	0.1 (0.01–0.5) <i>n</i> = 1 (<i>p</i> < 0.001)	0.03 (0.005–0.2) <i>n</i> = 1 (<i>p</i> = 0.008)	– <i>n</i> = 0	– <i>n</i> = 0
All, except Melanoma (<i>n</i> = 8,374)	0.8 (0.7–0.9) <i>n</i> = 658 (<i>p</i> < 0.001)	0.8 (0.8–0.9) <i>n</i> = 336 (<i>p</i> = 0.03)	0.8 (0.6–0.9) <i>n</i> = 94 (<i>p</i> = 0.02)	0.7 (0.6–0.8) <i>n</i> = 174 (<i>p</i> < 0.001)	0.6 (0.4–0.9) <i>n</i> = 22 (<i>p</i> = 0.01)	1.0 (0.71.4) <i>n</i> = 32 (<i>p</i> = 0.8)

^a Cox models with age, race, smoking, alcohol, body mass index (BMI), education, and marital status (versus Whites as referent). Separate models for Asians and specific Asian subsets ICD-9+ International classification of diseases, 9th Revision. UAD upper airway digestive. HR hazard ratio. CI 95 % confidence interval. *n* number

Table 5 Selected baseline traits of whites and Asian ethnicity groups

Trait	Whites	Chinese	Japanese	Filipino	South Asians	Other Asians
Male (%)	46.9 %	46.3 %	40.7	41.0 %	57.9 %	42.3 %
Smoke (%)	23.1 %	10.9 %	20.1 %	18.7 %	13.9 %	11.3 %
Smoke ≥ 1 pack/day	14.1 %	3.3 %	9.2 %	3.9 %	2.3 %	5.5 %
Never drank alcohol (%)	6.2 %	34.1 %	15.9 %	35.2 %	29.2 %	31.6 %
Drink ≥ 3 drinks/day	8.9 %	1.1 %	2.2 %	2.7 %	3.7 %	2.6 %
College graduate (%)	44.2 %	46.8 %	48.2 %	49.7 %	65.0 %	40.9 %
Mean BMI (kg/m ²)	24.1 kg/m ²	22.2 kg/m ²	22.8 kg/m ²	23.3 kg/m ²	23.4 kg/m ²	22.8 kg/m ²
BMI ≥ 30 kg/m ² (%)	7.6 %	1.5 %	2.4 %	3.2 %	3.0 %	3.9 %

Table 6 Hazard ratios (95 % confidence intervals) for selected covariates to risk of cancer*

Trait (referent)	All persons	Asians
Age ($\times 10$ years)	1.7 (1.7–1.7) ($p < 0.001$)	1.6 (1.6–1.7) ($p < 0.001$)
Male sex (female)	1.2 (1.1–1.2) ($p < 0.001$)	0.8 (0.8–0.9) ($p = 0.003$)
BMI ≥ 30 kg/m ² (≤ 25 kg/m ²)	1.1 (1.1–1.2) ($p < 0.001$)	1.4 (1.0–1.8) ($p = 0.02$)
Alcohol 1–2 drinks/day (never)	1.1 (1.1–1.2) ($p < 0.001$)	1.2 (1.1–1.6) ($p = 0.008$)
Alcohol ≥ 3 drinks/day (never)	1.2 (1.2–1.3) ($p < 0.001$)	1.3 (1.0–2.0) ($p = 0.04$)
Smoking < 1 ppd (never)	1.4 (1.3–1.4) ($p < 0.001$)	1.1 (1.0–1.3) ($p = 0.2$)
Smoking ≥ 1 ppd (never)	1.8 (1.7–1.9) ($p < 0.001$)	1.5 (1.1–1.8) ($p = 0.01$)
College graduate (no college)	1.0 (1.0–1.1) ($p = 0.2$)	1.1 (1.0–1.3) ($p = 0.1$)
Never married (married)	1.0 (1.0–1.1) ($p = 0.8$)	1.1 (0.9–1.3) ($p = 0.6$)

* Cox models with age, sex, smoking, alcohol, body mass index, education, and marital status

protection by increased epidermal melanin [9]. The UAD cancers are strongly related to smoking and alcohol drinking; thus, residual confounding by incomplete control for these traits is a possible factor in lower Asian risk. The lower risk of brain cancer, compared to Whites, was shared by Blacks (HR = 0.5, $p < 0.001$) and Hispanics (HR = 0.3, $p = 0.04$). An increased risk of brain cancer, particularly glioma, in Whites has been previously reported without identification of clear causal factors [28]. Possible explanations include access to care and disparate genetic mutations, e.g., p53 mutations and accumulation of tumor P53 protein in Whites with astrocytic glioma

[29] and the lower occurrence of CDKN2A/15 deletion in Japanese compared to White patients with malignant glioma [30].

Although lower risk of prostate cancer vs. Whites was substantial in all Asian ethnic groups, it remained the commonest cancer in Asian men. Lower risk of prostate cancer in Asians has been previously reported [31, 32] and remains unexplained. Lower Asian risk of bladder cancer, which we found only in men, also has been previously reported [33]. Lower risk of bladder cancer was also present in Blacks (HR = 0.6, $p < 0.001$) and Hispanics (HR = 0.4, $p = 0.001$). Smoking, chemicals, prior radiation or chemotherapy, chronic infection, and calculi are probable risk traits for bladder cancer; our analyses only controlled for smoking.

Published prospective data about risk of hematologic malignancies in US Asians are sparse. One report [34] suggests low incidence of non-Hodgkin's lymphoma. Another report [12] suggests that Asians have low risk of chronic lymphocytic leukemia and Hodgkin's disease but not of other malignancies in this group. A few reports [11, 12, 35] suggest lower risk in Asian countries. In our analysis, the lower risk of Asians was similar for several Asian ethnicities and for most hematologic malignancy subtypes except for multiple myeloma. It is noteworthy that our analysis confirmed the high risk of African Americans (HR = 2.4, $p < 0.001$) for this cancer type.

All cancer and Asian inter-ethnic differences

Since the lower overall cancer risk of Asians was driven largely by specific types (melanoma, prostate cancer, bladder cancer, and hematologic malignancies), we do not interpret the finding as indicating a universal low Asian cancer risk. In contrast with the disparities between the cancer types, differences between risks of Chinese, Japanese, and Filipinos do not seem large. South Asians are the exception since South Asians had substantially lower risk

than any other racial/ethnic group. While there are disparities between different South Asian populations [36, 37], there are previous reports [37–40] of low cancer risk in South Asians. Rastogi [32] speculates that a low cancer rate in India may be related to dietary factors, specifically widespread vegetarianism and use of the spice turmeric. This spice, used in curry powders, contains curcumin, which has antioxidant properties and, in animals, possible anticarcinogenic action [41, 42]. Goggins [39] and Wong [15] also mention possible dietary factors and also emphasize low smoking rates among South Asians as a probable factor. We do not have data about diet. South Asians in our study had a low prevalence of smoking (Table 6), but Chinese subjects were lower yet (Table 6). Obesity and heavy alcohol drinking may also be cancer risk traits, but several other groups in our population had lower prevalence of these traits than South Asians (Table 6). All of these were among the covariates in our models. Assuming a true lower cancer incidence among South Asians, an explanation is not evident.

To our knowledge, this is the first report showing that South Asians have lower cancer risk than any other race/ethnic group. It is noteworthy that these data about cancer in South Asians are a mirror-image opposite of the unexplained higher South Asian risk of coronary disease we have reported [43] in this study population. The higher coronary disease risk was also substantial vs. each other racial/ethnic group. However, despite the high incidence of coronary heart disease in this South Asian group, only 7 South Asians (~3 %) died of coronary disease. Thus, competing mortality before cancer incidence is an unlikely explanation for lower South Asian cancer risk. We speculate that these disparities in both coronary disease and cancer may reflect a greater genetic difference between South Asians and the other Asian groups.

Limitations

First is the fact that the demographics of US Asians have changed since our cohort was established. Our subjects were similar in age, sex, and race distribution to the general population in the area in 1978 to 1985 and reflected the socioeconomic spectrum of the general population except for the extremes of income [21]. However, there have been major changes in the demographics of Asian Americans over the last 30 years and our study population is relatively deficient in the number of some ethnic groups (e.g., South Asians, Koreans, and Vietnamese). Second is the fact that the examinees may have been more health conscious than the general population and, thus, perhaps less likely to develop certain cancer types. We think it unlikely that this aspect is systematically related to race–ethnicity. A third limitation is that selective retention in the health plan could

have affected results if it were systematically related to cancer risk by ethnicity, e.g., persons with cancer risk factors of a particular ethnicity might be more or less likely to remain members, but this type of retention bias seems unlikely to us. A fourth limitation is that selective completion of questionnaires related to lack of fluency in English could hypothetically have influenced the makeup of our study population. However, it seems unlikely that this was systematically related to cancer incidence. A fifth limitation is the absence of data about some cancer-specific risk factors and some general ones (e.g., diet or physical exercise, use of screening, or duration of US residence). A sixth limitation is the fact that South Asians were not self-classified as such. As we have detailed in Methods, we believe that our determination from other data was sensitive and specific. A seventh limitation is that data about habits and health measurements were obtained only at baseline. An eighth limitation is that small numbers for a number of cancer types limited the power of our analyses. This was especially important for South Asians.

Strengths

First, these prospective data with long follow-up and control for several potential confounders add substantially to previous very limited comprehensive reports about incident cancer in Asians. Second, there are only sparse previously published data comparing Asian ethnic groups to each other, and these data help fill that need. Finally, while there is considerable and appropriate interest in the role of accessibility to care and, in particular, cancer screening, as a factor in racial disparities in cancer incidence [17, 44, 45], we believe that such differences are an unlikely explanation in our comprehensive healthcare plan. Thus, equal access to care is a major strength of our analysis.

Conclusions

Cancer risks of Asians compared to White persons varied substantially among the cancer types, with evident increased, decreased, and similar risks. Melanoma contributed substantially to lower cancer risk among several Asian subgroups. Both environmental and genetic factors may be involved. Contrary to our expectations, inter-ethnic differences between Chinese, Japanese, and Filipinos were relatively few. Without apparent explanations, South Asians had lower overall cancer risk than each other ethnic group.

Acknowledgments The research was performed at the Division of Research of the Kaiser Permanente Medical Care Program, Oakland,

CA, with support by a Community Budget grant from the Kaiser Foundation Research Institute to Yan Li, MD, PhD as Principal Investigator. Data collection in 1978–1985 was supported by a grant to Dr. Arthur L. Klatsky from the Alcoholic Beverage Medical Research Foundation of Baltimore, MD. We are grateful to Cynthia Landy for assistance with data collection.

References

1. U.S. Census Bureau PD (2015) Annual estimates of the resident population by sex, race alone or in combination, and hispanic origin for the United States, states, and counties: April 1, 2010 to July 1, 2014
2. Kwong SL, Chen MS, Snipes KP et al (2005) Asian subgroups and cancer incidence and mortality rates in California. *Cancer* 104:2975–2981. doi:[10.1002/cncr.21511](https://doi.org/10.1002/cncr.21511)
3. McCracken M, Olsen M, Chen MS et al (2007) Cancer incidence, mortality, and associated risk factors among Asian Americans of Chinese, Filipino, Vietnamese, Korean, and Japanese ethnicities. *CA Cancer J Clin* 57:190–205. doi:[10.3322/canjclin.57.4.190](https://doi.org/10.3322/canjclin.57.4.190)
4. Bates JH, Hofer BM, Parikh-Patel A (2008) Cervical cancer incidence, mortality, and survival among Asian subgroups in California, 1990–2004. *Cancer* 113:2955–2963. doi:[10.1002/cncr.23752](https://doi.org/10.1002/cncr.23752)
5. Chen MS (2005) Cancer health disparities among Asian Americans: what we do and what we need to do. *Cancer* 104:2895–2902. doi:[10.1002/cncr.21501](https://doi.org/10.1002/cncr.21501)
6. Lauderdale DS, Huo D (2008) Cancer death rates for older Asian-Americans: classification by race versus ethnicity. *Cancer Causes Control* 19:135–146. doi:[10.1007/s10552-007-9079-4](https://doi.org/10.1007/s10552-007-9079-4)
7. Miller BA, Chu KC, Hankey BF, Ries L (2008) Cancer incidence and mortality patterns among specific Asian and Pacific Islander populations in the U.S. *Cancer Causes Control* 19:227–256. doi:[10.1007/s10552-007-9088-3](https://doi.org/10.1007/s10552-007-9088-3)
8. Huang V, Li W, Tsai J, Begier E (2013) Cancer mortality among Asians and Pacific Islanders in New York City, 2001–2010. *J Cancer Epidemiol* 2013:986408. doi:[10.1155/2013/986408](https://doi.org/10.1155/2013/986408)
9. Bradford PT Skin cancer in skin of color. *Dermatol Nurs* 21:170–7, 206; quiz 178
10. Meredith I, Sarfati D, Ikeda T, Blakely T (2012) Cancer in Pacific people in New Zealand. *Cancer Causes Control* 23:1173–1184. doi:[10.1007/s10552-012-9986-x](https://doi.org/10.1007/s10552-012-9986-x)
11. Carreon JD, Morton LM, Devesa SS et al (2008) Incidence of lymphoid neoplasms by subtype among six Asian ethnic groups in the United States, 1996–2004. *Cancer Causes Control* 19:1171–1181. doi:[10.1007/s10552-008-9184-z](https://doi.org/10.1007/s10552-008-9184-z)
12. Yang S-M, Li J-Y, Gale RP, Huang X-J (2015) The mystery of chronic lymphocytic leukemia (CLL): why is it absent in Asians and what does this tell us about etiology, pathogenesis and biology? *Blood Rev* 29:205–213. doi:[10.1016/j.blre.2014.12.001](https://doi.org/10.1016/j.blre.2014.12.001)
13. Gomez SL, Noone AM, Lichtensztajn DY et al (2013) Cancer incidence trends among Asian American populations in the United States, 1990–2008. *J Natl Cancer Inst* 105:1096–1110. doi:[10.1093/jnci/djt157](https://doi.org/10.1093/jnci/djt157)
14. Chang ET, Keegan THM, Gomez SL et al (2007) The burden of liver cancer in Asians and Pacific Islanders in the Greater San Francisco Bay Area, 1990 through 2004. *Cancer* 109:2100–2108. doi:[10.1002/cncr.22642](https://doi.org/10.1002/cncr.22642)
15. Wong R, Corley DA (2008) Racial and ethnic variations in hepatocellular carcinoma incidence within the United States. *Am J Med* 121:525–531. doi:[10.1016/j.amjmed.2008.03.005](https://doi.org/10.1016/j.amjmed.2008.03.005)
16. Park S, Bae J, Nam B-H, Yoo K-Y (2008) Aetiology of cancer in Asia. *Asian Pac J Cancer Prev* 9:371–380
17. Wang JH, Sheppard VB, Schwartz MD et al (2008) Disparities in cervical cancer screening between Asian American and Non-Hispanic white women. *Cancer Epidemiol Biomarkers Prev* 17:1968–1973. doi:[10.1158/1055-9965.EPI-08-0078](https://doi.org/10.1158/1055-9965.EPI-08-0078)
18. Tran HN, Li Y, Siu S et al (2013) Predictors of lung cancer: noteworthy cell type differences. *Perm J* 17:23–29. doi:[10.7812/TPP/12-104](https://doi.org/10.7812/TPP/12-104)
19. Taylor VM, Nguyen TT, Jackson JC, McPhee SJ (2008) Cervical cancer control research in Vietnamese American communities. *Cancer Epidemiol Biomarkers Prev* 17:2924–2930. doi:[10.1158/1055-9965.EPI-08-0386](https://doi.org/10.1158/1055-9965.EPI-08-0386)
20. Klatsky AL, Siegelau AB, Landy C, Friedman GD (1983) Racial patterns of alcoholic beverage use. *Alcohol Clin Exp Res* 7:372–377
21. Krieger N (1992) Overcoming the absence of socioeconomic data in medical records: validation and application of a census-based methodology. *Am J Public Health* 82:703–710
22. Collen MF, Davis LF (1969) The multitest laboratory in health care. *J Occup Med* 11:355–360
23. McCracken M, Olsen M, Chen MS, et al. Cancer incidence, mortality, and associated risk factors among Asian Americans of Chinese, Filipino, Vietnamese, Korean, and Japanese ethnicities. *CA Cancer J Clin* 57:190–205
24. Goh L-Y, Leow AHR, Goh K-L (2014) Observations on the epidemiology of gastrointestinal and liver cancers in the Asia-Pacific region. *J Dig Dis* 15:463–468. doi:[10.1111/1751-2980.12164](https://doi.org/10.1111/1751-2980.12164)
25. Nguyen EV (2003) Cancer in Asian American males: epidemiology, causes, prevention, and early detection. *Asian Am Pac Isl J Health* 10:86–99
26. Choi CE, Sonnenberg A, Turner K, Genta RM (2015) High prevalence of gastric preneoplastic lesions in East Asians and hispanics in the USA. *Dig Dis Sci* 60:2070–2076. doi:[10.1007/s10620-015-3591-2](https://doi.org/10.1007/s10620-015-3591-2)
27. Cortes MCC, Yamakawa A, Casingal CR et al (2010) Diversity of the cagA gene of *Helicobacter pylori* strains from patients with gastroduodenal diseases in the Philippines. *FEMS Immunol Med Microbiol* 60:90–97. doi:[10.1111/j.1574-695X.2010.00722.x](https://doi.org/10.1111/j.1574-695X.2010.00722.x)
28. Wrensch M, Minn Y, Chew T et al (2002) Epidemiology of primary brain tumors: current concepts and review of the literature. *Neuro Oncol* 4:278–299
29. Chen P, Aldape K, Wiencke JK et al (2001) Ethnicity delineates different genetic pathways in malignant glioma. *Cancer Res* 61:3949–3954
30. Mochizuki S, Iwade Y, Namba H et al (1999) Homozygous deletion of the p16/MTS-1/CDKN2 gene in malignant gliomas is infrequent among Japanese patients. *Int J Oncol* 15:983–989
31. Raymundo EM, Rice KR, Chen Y et al (2011) Prostate cancer in Asian Americans: incidence, management and outcomes in an equal access healthcare system. *BJU Int* 107:1216–1222. doi:[10.1111/j.1464-410X.2010.09685.x](https://doi.org/10.1111/j.1464-410X.2010.09685.x)
32. Kimura T (2012) East meets West: ethnic differences in prostate cancer epidemiology between East Asians and Caucasians. *Chin J Cancer* 31:421–429. doi:[10.5732/cjc.011.10324](https://doi.org/10.5732/cjc.011.10324)
33. Anton-Culver H, Lee-Feldstein A, Taylor TH (1993) The association of bladder cancer risk with ethnicity, gender, and smoking. *Ann Epidemiol* 3:429–433
34. Herrinton LJ, Goldoft M, Schwartz SM, Weiss NS (1996) The incidence of non-Hodgkin's lymphoma and its histologic subtypes in Asian migrants to the United States and their descendants. *Cancer Causes Control* 7:224–230
35. Soerjomataram I, Lortet-Tieulent J, Parkin DM, et al. (2012) Global burden of cancer in 2008: a systematic analysis of disability-adjusted life-years in 12 world regions. *Lancet (London, England)* 380:1840–50. doi:[10.1016/S0140-6736\(12\)60919-2](https://doi.org/10.1016/S0140-6736(12)60919-2)

36. Hislop GT, Bajdik CD, Regier MD, Barroetavena MC (2007) Ethnic differences in survival for female cancers of the breast, cervix and colorectum in British Columbia, Canada. *Asian Pac J Cancer Prev* 8:209–214
37. Moore MA, Manan AA, Chow KY et al (2010) Cancer epidemiology and control in peninsular and island South-East Asia—past, present and future. *Asian Pac J Cancer Prev* 11(Suppl 2):81–98
38. Rastogi T, Devesa S, Mangtani P et al (2008) Cancer incidence rates among South Asians in four geographic regions: India, Singapore, UK and US. *Int J Epidemiol* 37:147–160. doi:[10.1093/ije/dym219](https://doi.org/10.1093/ije/dym219)
39. Goggins WB, Wong G (2009) Cancer among Asian Indians/Pakistanis living in the United States: low incidence and generally above average survival. *Cancer Causes Control* 20:635–643. doi:[10.1007/s10552-008-9275-x](https://doi.org/10.1007/s10552-008-9275-x)
40. Jin H, Pinheiro PS, Xu J, Amei A (2016) Cancer incidence among Asian American populations in the United States, 2009–2011. *Int J Cancer* 138:2136–2145. doi:[10.1002/ijc.29958](https://doi.org/10.1002/ijc.29958)
41. Surh Y-J, Chun K-S (2007) Cancer chemopreventive effects of curcumin. *Adv Exp Med Biol* 595:149–172. doi:[10.1007/978-0-387-46401-5_5](https://doi.org/10.1007/978-0-387-46401-5_5)
42. Tayyem RF, Heath DD, Al-Delaimy WK, Rock CL (2006) Curcumin content of turmeric and curry powders. *Nutr Cancer* 55:126–131. doi:[10.1207/s15327914nc5502_2](https://doi.org/10.1207/s15327914nc5502_2)
43. Hajra A, Li Y, Siu S et al (2013) Risk of coronary disease in the South Asian American population. *J Am Coll Cardiol* 62:644–645. doi:[10.1016/j.jacc.2013.05.048](https://doi.org/10.1016/j.jacc.2013.05.048)
44. Kandula NR, Wen M, Jacobs EA, Lauderdale DS (2006) Low rates of colorectal, cervical, and breast cancer screening in Asian Americans compared with non-Hispanic whites: cultural influences or access to care? *Cancer* 107:184–192. doi:[10.1002/cncr.21968](https://doi.org/10.1002/cncr.21968)
45. Laiyemo AO, Doubeni C, Pinsky PF et al (2010) Race and colorectal cancer disparities: health-care utilization vs different cancer susceptibilities. *J Natl Cancer Inst* 102:538–546. doi:[10.1093/jnci/djq068](https://doi.org/10.1093/jnci/djq068)